SEAN HAYES
Distributed Systems & Backend Engineer (Kubernetes • AWS • Kafka • High-Scale Reliability)
Active Top Secret Clearance (Activated 08/01/2023)
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SUMMARY

Distributed systems and platform engineer with 4+ years of experience building and operating cloud-native services at scale. Strong background in designing resilient, event-driven architectures using Kafka, AWS, and Kubernetes. Proven track record in production ownership, on-call rotations, incident response, and improving system reliability through observability, automation, and fault-tolerant design. Passionate about high-throughput systems, real-time processing, and defending large-scale platforms through robust engineering practices. Currently deepening expertise in Kubernetes networking, Elastic, and distributed system performance.

CORE SKILLS

- Distributed Systems, Event-Driven Architecture, High Availability, Fault Tolerance
- Kubernetes, Helm, Docker, EKS
- Java (Spring Boot), Python, Go (foundational)
- AWS: Lambda, S3, SQS, DynamoDB, API Gateway, CloudWatch, IAM
- Messaging & Storage: Kafka, Redis, PostgreSQL
- Observability: Elastic/ELK, Prometheus, Grafana
- CI/CD & Quality: GitHub Actions, GitLab CI/CD, SonarQube, dependency scanning
- Infrastructure as Code: Terraform
- Networking fundamentals: HTTP, TCP/IP, load balancing, API gateways
- Practices: On-call, incident response, postmortems, resilience engineering, security and compliance

PROFESSIONAL EXPERIENCE

Duke Energy – Full Stack Engineer
Charlotte, NC | Apr 2026 – Present

- Develop and maintain event-driven distributed systems integrating Kafka, AWS Lambda, and Azure SQL to automate enterprise work order processing.
- Design Python-based Kafka consumer services using AWS Lambda Self-Managed Pollers, AWS Secrets Manager, and HashiCorp Vault for secure authentication and message processing.
- Build resilient message-processing pipelines that consume AVRO-serialized events and transform them into downstream relational data models.
- Provision and manage cloud infrastructure using Terraform and HCP Terraform, with GitHub Actions workflows automating infrastructure plans and deployments.
- Implement infrastructure-as-code for Lambda functions, IAM roles, networking, secrets management, and event-driven integrations.

• Troubleshoot distributed messaging, authentication, networking, and infrastructure issues across Kafka brokers, Lambda event source mappings, Vault, and cloud services.
• Contribute TypeScript enhancements supporting internal workflow automation and operational tooling.
• Collaborate with platform, infrastructure, and data engineering teams to deliver secure, highly available integration pipelines across AWS and Azure.

General Dynamics Information Technology (GDIT) – Software Engineer III
Charlotte, NC | Jan 2023 – Jan 2025

• Designed, deployed, and operated containerized Spring Boot microservices on Kubernetes using Helm, improving deployment safety and service reliability across distributed environments.
• Built and maintained GitLab CI/CD pipelines with SonarQube quality gates, increasing deployment confidence and reducing regressions.
• Collaborated with security and platform teams to implement least-privilege access controls and secure cloud-native service architectures.
• Participated in on-call rotations, triaged production incidents, and performed root cause analysis to improve system reliability and reduce recurring issues.
• Led code reviews, design discussions, and mentoring efforts that promoted engineering best practices across the team.

TIAA – Software Engineer (Backend / Cloud)
Charlotte, NC | Jan 2021 – Dec 2022

• Designed and implemented event-driven microservices and Kafka pipelines supporting high-volume asynchronous data processing.
• Built resilient distributed systems using dead-letter queues, idempotent processing, and backpressure strategies to improve fault tolerance under peak load.
• Integrated AWS services including Lambda, S3, SQS, DynamoDB, IAM, and Secrets Manager into secure, serverless application workflows.
• Built and optimized serverless data pipelines using AWS Glue, Athena, and Redis caching to improve query performance and application responsiveness.
• Improved system observability through ELK instrumentation, custom alerting, post-incident reviews, and permanent reliability improvements that reduced MTTR.

EDUCATION & CERTIFICATIONS

University of North Carolina at Charlotte (2020) – B.S. Computer Science (AI & Robotics) Magna Cum Laude

Certifications:
• AWS Certified Cloud Practitioner
• Google Data Analytics Professional Certificate